

Glove Vocaliser Project Report

Project Title: Glove Vocaliser

Group Number: 34

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Mentor: Prof.P.M. Jat

Course Number:PC 122

1. Introduction

The **Hand Vocaliser** is an innovative wearable system that detects hand gestures and translates them into digital actions such as game controls or speech designed **with** high-accuracy flex sensors **and a** combined inertial measurement unit (IMU), it provides an intuitive and accessible interface for both entertainment and assistive technologies. This system can help bridge the communication gap for speech-impaired individuals and create immersive experiences for gamers and VR users.

2. Motivation

Traditional communication aids for the speech-impaired are often expensive and lack real-time responsiveness. Similarly, gesture-based interfaces in gaming and VR are usually high-end and unaffordable for many. This project is motivated by the desire to develop a cost-effective, real-time wearable solution that enhances accessibility and human-computer interaction through natural hand gestures

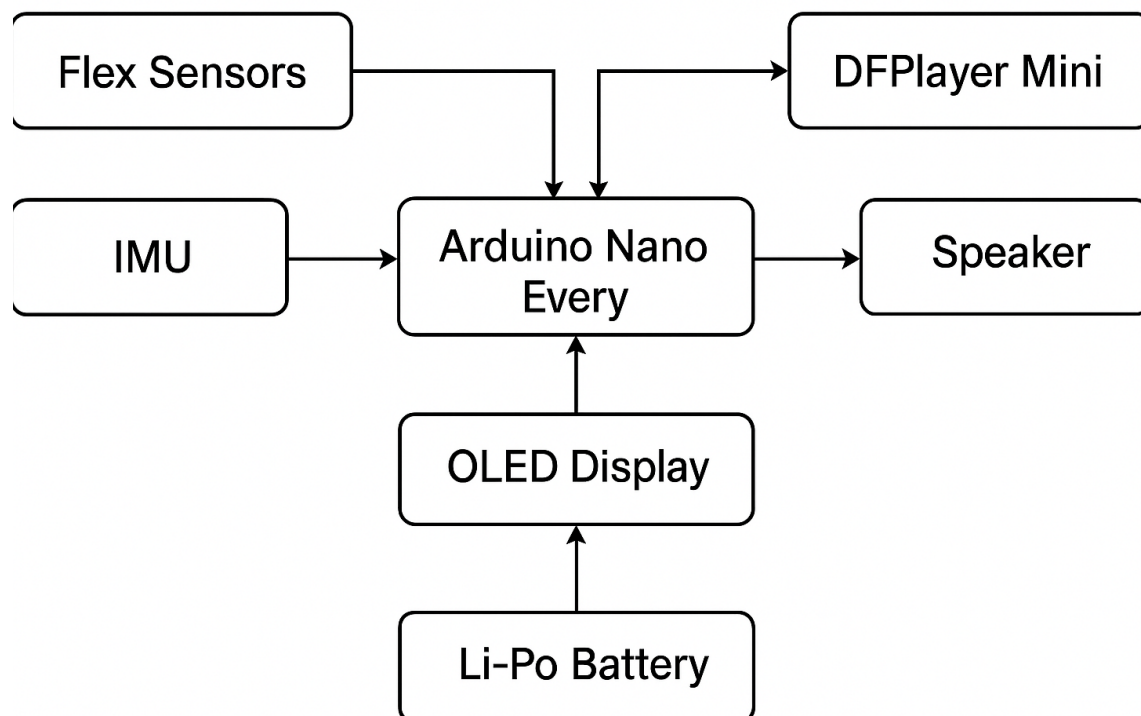
3. Proposed Solution

The Glove Vocaliser uses flex sensors on each finger to detect bends and a BNO055 IMU to measure hand orientation. Data is collected by an Arduino Nano Every. After processing, it triggers voice outputs via a DFPlayer Mini module connected to a speaker. The glove is powered by a rechargeable Li-Po battery and includes an OLED display for gesture feedback (optional).

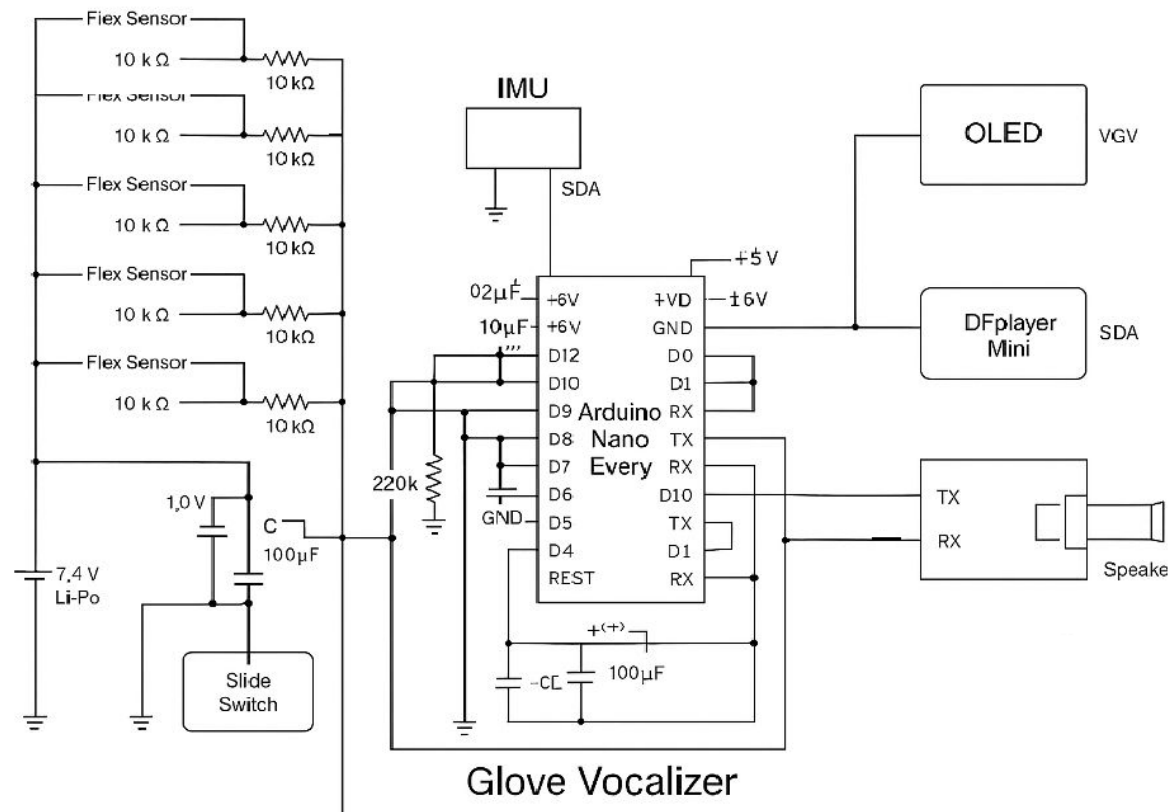
Major Components:

- Arduino Nano Every
- Flex Sensors (5x)
- BNO055 IMU
- DFPlayer Mini + Speaker
- OLED Display (optional)
- Li-Po Battery with TP4056 Charger
- AMS1117 5V Voltage Regulator

Block Diagram:



Circuit Diagram:



Circuit Overview:

1. Microcontroller

Arduino Nano Every is the brain of the system.

- Reads data from sensors (Flex + IMU).
- Processes the data and sends output signals to the DFPlayer Mini and OLED Display (optional).

2. Input Sensors

a. Flex Sensors (5 units)

- Placed on each finger.
- Output analog resistance based on finger bending.
- Connected to analog input pins (A0–A4) of Arduino.
- Paired with 10kΩ resistors in a voltage divider configuration.

b. IMU (BNO055)

- Connected via I2C interface (SDA & SCL pins) to Arduino.
 - Provides orientation and motion data to refine gesture detection.
 - Internally fuses accelerometer, gyroscope, and magnetometer data.
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3. Audio Output

DFPlayer Mini Module

- Receives Serial commands from Arduino (using SoftwareSerial).
 - Reads pre-recorded MP3 files from an SD card.
 - Plays voice/audio output through a connected 3W speaker.
 - Connections:
 - TX/RX of Arduino (SoftwareSerial)
 - 5V and GND
 - Speaker connected via SPK1/SPK2 pins.
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4. OLED Display (Optional)

0.96" OLED (I2C)

- Displays the detected gesture or system status in real-time.
 - Connected to I2C lines (SDA/SCL) shared with BNO055.
 - Libraries like Adafruit_SSD1306 or U8g2 can be used for the display.
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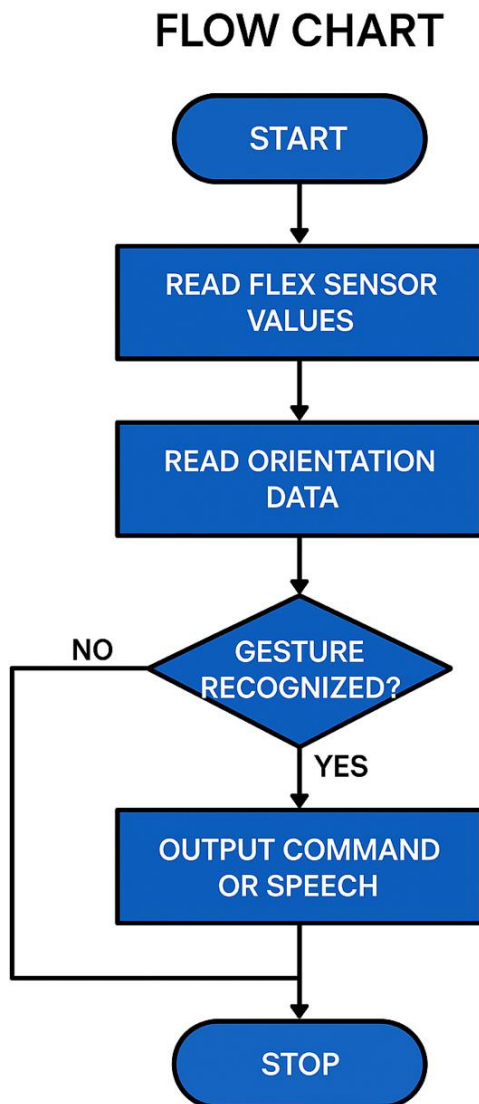
5. Power Supply

- Powered by a **7.4V Li-Po battery**.
 - Battery connected to a **TP4056 charging module** with protection circuit.
 - A **voltage regulator** (AMS1117 5V or a buck converter) provides stable 5V for Arduino and modules.
 - ON/OFF control via a slide switch.
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6. Additional Components

- **Wireless Module (NRF24L01 / BLE):** Enables wireless transmission of recognized gestures or system status to external devices like a mobile app or PC. Useful for portability and remote control.
- **EEPROM or MicroSD Module:** Used to store gesture logs, audio mappings, or configuration data. Allows future upgrade to dynamic gesture training and offline data analysis.
- **220Ω resistors** for status/power LEDs.
- **0.1μF capacitors** for noise filtering on power lines.
- **Heat shrink tubing, Velcro straps, and tape** for securing sensors and electronics to the glove.

Flowchart:



4. Timeline (PC 223 Plan)

1.SUMMER VACATION

In this time we will do all the necessary research for implementing the project and search for the materials required.

2.END OF JULY

By this time the 3rd sem would have started and we would be back on campus, and discuss all the research we did with our mentor

3.MID AUGUST

By this time we expect to have all the necessary materials and start working on the project.

4.MID SEPTEMBER

We expect to have made a working prototype by this time and would start converting into final model.

5.START OF OCTOBER

We have kept this time to solve any problems faced or to improve our final model as per the mentors guidance..

6.MID OF OCTOBER

This period serves as a buffer in case of unforeseen delays.

7.SUBMISSION

Submission of the final working model of our project in front of evaluation panel.

5. Budget and Justification

S. No	Component	Specification	Qty	Unit Price (Rs.)	Total (Rs.)
1	Arduino Nano Every	ATmega4809, compact MCU	1 No	500	500
2	Flex Sensors (2.2")	Spectra Symbol, bend detection	5 Nos	400	2000
3	BNO055 IMU	9-axis with onboard fusion	1 No	1200	1200
4	DFPlayer Mini + Speaker	MP3 voice playback module	1 No	400	400
5	OLED Display (Optional)	0.96", I2C	1 No	180	180
6	Li-Po Battery + Charger	7.4V 2S, TP4056 module	1 Set	500	500
7	Voltage Regulator (AMS1117)	5V output	1 No	30	30
8	Resistors (10k Ω , 220 Ω)	Voltage divider + LED protection	15 Nos	1	15
9	Capacitors (0.1 μ F, 100 μ F)	Noise filtering	7 Nos	2	14
10	Slide Switch	SPDT switch for power control	1 No	10	10
11	Wires, Connectors, PCB	Assembly materials	1 Set	200	200
12	Glove	Stretchable base for sensors	1 No	200	200
13	Wireless Module	NRF24L01 / BLE	1 No	300	300
14	EEPROM or MicroSD Module	Storage for gesture mapping/logs	1 No	200	200

Component	Specification	Quantity
Resistor	10kΩ, 1/4W, ±1%	10
Resistor	220Ω, 1/4W, ±1%	10
Capacitor	0.1μF (ceramic disc)	5
Capacitor	100μF, 25V (electrolytic)	2
LED	5mm Red	3
LED	5mm Green	2
Slide Switch	SPDT	1
Male-Female Jumper Wires	20cm	20
Screw Terminals	2-pin	5
Heat Shrink Tubing	Assorted Sizes	1 set
PCB (Custom or general)	Single/Double layer	1
Mounting/Velcro Tape	For glove fixing	1 roll
Gloves (cotton/flex)	Wearable base	1 pair

Total Estimated Cost: Rs. 5749 – 6000.